Three-Dimensional Choreographic Orbits: A Point-Group Classification Programme

Research Design Document — v0.2

e5bed97b3349 • March 2026

Abstract

We propose a systematic programme to discover, classify, and characterise three-dimensional equal-mass choreographic solutions to the Newtonian N-body problem. A *choreographic orbit* is a periodic solution in which all N bodies traverse a single closed curve, equally spaced in phase. While the two-dimensional case has been surveyed extensively — most notably by Simó (2002) — fully three-dimensional choreographies remain largely unexplored. Our programme organises the search space by the three-dimensional point-group symmetry of the orbit, imports the classification machinery of molecular symmetry into gravitational dynamics, and characterises each discovered orbit by the knot type of its spacetime trace. We expect the resulting catalogue to reveal a rich correspondence between symmetry class, knot invariant, and action value, and to draw explicit connections to the emerging literature on mechanically synthesised molecular knots.

1 Background and Motivation

1.1 The choreographic orbit

Consider N bodies of equal mass m moving in $\mathbb{R}^3$ under mutual Newtonian gravity. A *choreographic solution* is a periodic orbit $\mathbf{q}_i(t) \in \mathbb{R}^3$ ($i = 1, \ldots, N$, period T) such that there exists a single closed curve $\gamma \subset \mathbb{R}^3$ and a phase shift $\tau = T/N$ with

$$\mathbf{q}_i(t) = \mathbf{q}_1(t + (i-1)\,\tau)\,, \quad i = 1, \ldots, N.$$

All bodies follow the same spatial curve; they are simply offset in time. The solution is therefore entirely specified by the single curve γ and its parameterisation.

1.2 Historical context

The existence of non-trivial choreographic solutions was not established rigorously until 2000, when Chenciner and Montgomery [1] proved the existence of the figure-eight solution for $N = 3$ by showing it minimises the Lagrangian action over a

topologically constrained loop space. This variational proof transformed the problem: rather than searching phase space by continuation or shooting, one seeks *minima of an action functional subject to symmetry constraints*.

Simó [2] immediately exploited this framework numerically, discovering hundreds of planar ($\mathbb{R}^2$) choreographies for small N and organising them by their discrete symmetry groups in the plane. This catalogue constitutes the current state of the art for the two-dimensional case. Subsequent work has extended the analysis to figure-eight stability [3], choreographies with non-equal masses, and related classes of symmetric periodic orbits (hip-hop solutions, p/q-choreographies).

1.3 The three-dimensional gap

The extension to genuine three-dimensional choreographies — orbits that are not merely planar solutions embedded in $\mathbb{R}^3$ — is conspicuously underdeveloped. The obstacles are partly technical (the symmetry group is richer and harder to parameterise) and partly conceptual (it is less obvious which topological sectors of the loop space admit minimisers). The companion Mathematical Framework document [7] resolves the conceptual question: Propositions 4.1–4.3 and the summary table in that document determine *precisely* which three-dimensional point groups admit genuinely non-planar choreographies and which force the orbit into a plane. Non-planarity is not an empirical question to be settled numerically; it is a structural consequence of the representation theory of G on the Fourier coefficient space. We identify the systematic search over the non-planar symmetry classes as the central programme motivating this work.

1.4 Chemistry as organising principle

The symmetry group of a three-dimensional choreographic orbit is a subgroup of $O(3)$ acting on the curve γ. These are precisely the *point groups* familiar from molecular symmetry: C_n, C_{nv}, C_{nh}, D_n, D_{nh}, D_{nd}, T, T_d, T_h, O, O_h, I, I_h. The Schönflies classification, originally developed to organise molecular geometry, provides a ready-made taxonomy for three-dimensional choreographies. Orbits with high symmetry (icosahedral I_h, octahedral O_h) impose strong constraints on the curve γ that dramatically reduce the search space and may guarantee the existence of minimisers via symmetric criticality arguments [4].

A second bridge to chemistry is topological. The spacetime trace of a choreographic orbit — the curve $t \mapsto (\mathbf{q}_1(t), t)$ projected back to $\mathbb{R}^3$ after closing the time axis — is a closed curve whose knot type is a topological invariant of the orbit. Molecular knot synthesis (trefoil knots, Solomon links, torus knots) has undergone rapid development [6]; a gravitational analogue in which orbit topology is controlled by point-group symmetry would constitute a meaningful conceptual contribution to both fields.

2 Mathematical Framework

2.1 Action functional and choreographic constraint

The Lagrangian action over one period is

$$\mathcal{A}[\mathbf{q}_1, \ldots, \mathbf{q}_N] \;=\; \int_0^T \mathcal{L}\, dt \;=\; \int_0^T \left[\sum_{i=1}^{N} \tfrac{m}{2} \, |\dot{\mathbf{q}}_i|^2 \;+\; \sum_{i<j} \frac{Gm^2}{|\mathbf{q}_i - \mathbf{q}_j|} \right] dt.$$

Restricting to the choreographic subspace via $\mathbf{q}_i(t) = \mathbf{q}_1(t + (i-1)T/N)$ reduces the problem to a single-curve functional

$$\mathcal{A}[\gamma] \;=\; \frac{N}{T} \int_0^T \left[\tfrac{m}{2} |\dot{\gamma}|^2 + \frac{1}{N} \sum_{k=1}^{N-1} \frac{Gm^2}{|\gamma(t) - \gamma(t + kT/N)|} \right] dt,$$

where $\gamma : [0, T] \to \mathbb{R}^3$ is T-periodic. Critical points of $\mathcal{A}$ on the appropriate Sobolev space $H^1(S^1, \mathbb{R}^3)$ correspond to choreographic solutions. Global minimisers in a given homotopy class exist whenever collisions can be excluded from that class [1, 5].

2.2 Symmetry constraints and reduction

Let $G \subset O(3) \times \mathbb{Z}/N\mathbb{Z}$ be the symmetry group of the orbit, acting on the loop space by simultaneously rotating the curve and permuting the bodies. By the Principle of Symmetric Criticality [4], a symmetric critical point of $\mathcal{A}|_{\mathrm{Fix}(G)}$ is a critical point of the full action. This reduces the search to the fixed-point subspace $\mathrm{Fix}(G)$, whose dimension is determined by the representation theory of G on $H^1(S^1, \mathbb{R}^3)$.

Concretely, for a C_{3v} orbit with $N = 3$ bodies, the constraint forces γ to pass through specific planes and axes, yielding a finite-dimensional initial-condition manifold to search. For icosahedral symmetry with $N = 12$ or $N = 60$, the constraints are far stronger and the search space dramatically smaller.

2.3 Knot theory of the spacetime trace

Define the *spacetime trace* of the choreography as the closed curve

$$\Gamma : t \mapsto \Big(\mathbf{q}_1(t),\; t \bmod T \Big) \;\in\; \mathbb{R}^3 \times S^1.$$

The isotopy class of Γ in the complement of the other bodies' spacetime traces is a topological invariant. For planar choreographies this reduces to a braid word in B_N; for three-dimensional ones it is a richer object, potentially a knot or link in S^3. We will compute knot invariants — at minimum the Alexander polynomial and the HOMFLY-PT polynomial — for each discovered orbit.

2.4 Stability

Linear stability is assessed via Floquet theory: we compute the monodromy matrix $M = \Phi(T)$ of the linearised variational equations along the orbit and classify eigenvalues as elliptic (on the unit circle), hyperbolic (real), or complex-hyperbolic. An orbit is *linearly stable* if all eigenvalues lie on the unit circle. We will also compute the first few Lyapunov exponents numerically to characterise the nonlinear stability basin.

3 Research Questions

Research Question 1. For each three-dimensional point group G (Schönflies classification), does there exist a choreographic N-body orbit with symmetry group containing G? For which N and G pairs do minimisers exist in the relevant homotopy class?

Research Question 2. What is the knot type of the spacetime trace of each discovered orbit? Is there a systematic map from symmetry class to knot invariant?

Research Question 3. What is the stability character (elliptic, hyperbolic, complex-hyperbolic) of each discovered orbit? Are high-symmetry orbits generically more or less stable than low-symmetry ones?

Research Question 4. Do the action values $\mathcal{A}^\star(G, N)$ at minimisers admit a closed-form or asymptotic expression in the symmetry order $|G|$ and body count N?

Research Question 5. Is there a formal correspondence between orbits whose spacetime traces realise specific knot types and known or hypothetical molecular knot topologies?

4 Methodology

4.1 Phase 1: Symmetry-constrained search (N = 3, 4)

For each point group G compatible with small N, we:

1. Parameterise the fixed-point subspace $\mathrm{Fix}(G)$ using a Fourier truncation $\gamma(t) \approx \sum_{k=0}^{K}(a_k \cos(2\pi kt/T) + b_k \sin(2\pi kt/T))$ with symmetry-imposed linear constraints on the coefficients $a_k, b_k \in \mathbb{R}^3$.
2. Minimise $\mathcal{A}$ over the truncated space using gradient descent with an adaptive step (L-BFGS), starting from multiple random initialisations within the symmetry-compatible manifold.
3. Refine converged candidates using a symplectic integrator (Yoshida 6th-order or Forest–Ruth) and Newton–Raphson correction to a true periodic orbit.
4. Discard near-collisions and planar solutions (those for which γ lies within ε of a fixed plane).

4.2 Phase 2: Higher symmetry and larger N

Extend to $N = 6, 12$ with tetrahedral and octahedral symmetry groups, exploiting the stronger constraints to keep the search space tractable. Extend further to icosahedral symmetry at $N = 12$ and $N = 60$. The *dimensional miracle* established in MathFrame [7] §4.6 shows that the search-space dimension at fixed truncation ratio K/N is independent of the group order $|G|$: icosahedral constraints impose no greater computational burden than tetrahedral ones. This formally justifies the icosahedral case as a primary target rather than a speculative extension.

4.3 Phase 3: Knot classification

For each discovered orbit, compute:

- The writhe and crossing number of Γ from a projection.
- The Alexander polynomial $\Delta(t)$ via the Seifert matrix.
- The HOMFLY-PT polynomial $P(v, z)$ as a stronger invariant.
- The hyperbolic volume (if Γ is hyperbolic) via SnapPy.

4.4 Phase 4: Stability analysis

Integrate the variational equations alongside the orbit to obtain the monodromy matrix. Compute eigenvalues and classify stability. For marginally stable orbits, explore KAM tori numerically via surface-of-section.

4.5 Phase 5: Visualisation

Produce publication-quality renders: orbital traces as tubes in $\mathbb{R}^3$ with point-group symmetry axes shown explicitly, spacetime braids rendered as three-dimensional knot diagrams, and high-fidelity video renders (4K/60fps, H.264) of the full choreographic motion for public dissemination.

5 Expected Outputs

1. **A catalogue of 3D choreographies**, organised by $(N, G, \text{knot type})$, with action values, stability characters, and initial conditions provided to full double precision.
2. **A correspondence table** mapping symmetry class to knot invariant, with conjectures about which cells are empty and why.
3. **A computational toolkit** (Python, open-source) for reproducing all results and searching for new orbits.
4. **A research paper** suitable for submission to a journal such as *Celestial Mechanics and Dynamical Astronomy*, *Nonlinearity*, or *Communications in Mathematical Physics*.
5. **A visual portfolio** for public dissemination across scientific social media.

6 Dissemination Plan

The work will be released in two stages. The catalogue and toolkit will be posted simultaneously to arXiv (math-ph / math.DS) and GitHub under a permissive open-source licence, with DOI minting via Zenodo to ensure citability. The paper will be submitted to a peer-reviewed journal concurrently with the arXiv preprint.

Public communication will begin with the arXiv release and proceed across a coordinated set of accounts on X, Bluesky, Farcaster, and Hacker News, accompanied by the animated visualisations. Given the visual character of the work and the chemistry connection, we anticipate organic reach into both the physics and chemistry communities.

7 Open Questions and Risks

- **Collision avoidance**: Minimisers in some symmetry classes may pass through collisions, rendering the action infinite or the orbit non-physical. Excluding such classes analytically or verifying their emptiness numerically is a prerequisite for clean results.
- **Uniqueness**: Multiple minimisers may exist within a single symmetry-homotopy class. The relationship between connected components of $\mathrm{Fix}(G)$ and distinct orbit families is not understood a priori.
- **Fourier truncation error**: Highly irregular curves (low-action orbits near collision) may require large K for accurate representation. Convergence monitoring will be required.
- **Genuinely new orbits**: There is a possibility — small but non-negligible — that 3D choreographies simply do not exist beyond degenerate (planar or hip-hop) cases for small N. A null result, reported carefully, would itself be a contribution.

References

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